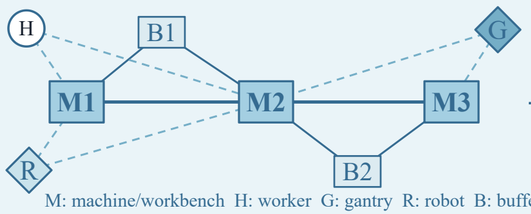
(a) Heterogeneous resource representation



M: machine/workbench H: worker G: gantry R: robot B: buffer

Graph G
M^G, E_lap

Node-aligned history

X: 30 × 38 × 27

6 groups

Grouped Linear + GELU
Mean aggregation

Time + Laplacian
+ node embeddings

(b) Process-informed supervision and pattern clustering

Operational evidence

Queue | Stall / shortage | Logistics

Resource-specific rules + persistence

Raw

Sustained

Event / state targets

Cause targets

Supervision

History-derived conditioning

Rule-based regime IDs → **Regime embedding**

Local + neighbour cues → **Precursor MLP**

Short-pattern clustering

Activity patches
(length 3)

k-means

16 prototypes P

PHGST-DP

$H^{(0)}$

Encoder block × 5

LayerNorm

Temporal
4 heads

Geographic
2 heads, M^G

Semantic
2 heads, M^s

Concat + Linear

residual

LayerNorm

Feed-forward network

residual

$H(t)$

Regime + precursor fusion (b)

Skip + order-progress fusion

Graph pooling + node evidence

Event heads

State / remaining time decoders

Process-cause classifier

Training loss

Event decoding and report assembly

(c) Delay-conditioned interaction

Earlier precursor

Later constraint

lag

Pattern matching

Queue query patch

Activity prototypes P (b)

Q^P 38 × 20

K^P 16 × 20

V^P 16 × 20

$\alpha^P = \mathrm{softmax}(Q^P K^{PT} / \sqrt{d_h})$

Pattern weights 38 × 16

Pattern context C^P
38 × 20

Geographic key update

$\tilde{K}^G = K^G + C^P$

Base key K^G + Context C^P → **Updated key $\tilde{K}^G$**

Q^G, V^G → $\mathrm{softmax}(Q^G \tilde{K}^{GT} / \sqrt{d_h} + M^G) V^G$

Structured bottleneck report

Resource i*

t | $t + \hat{s}$ | $t + \hat{s} + \hat{d}$

State

Onset $\hat{s}$

Occurrence $\hat{w}$

Duration $\hat{d}$

Process cause $\hat{c}$

Order remaining time $\hat{\ell}$